Assignment_2

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1. Complete Exercise 6.5 in Hsieh's book. Please build an MLP NN model and use the cross-validation technique to tune at least one model hyperparameter other than the learning rate.

> **6.5**
>
> Build a non-linear regression model for predicting the winter snow water equivalent (SWE) for the dataset provided on the book website (same dataset as used in Exercise 5.8), with the Niño 3.4 and the PNA climate indices as the two predictors. Use only the first 40 years of data to build the model, reserving the remaining 28 years as independent test data. Compute the RMSE and Pearson correlation between the observed and predicted SWE values over the test data. Compare these performance scores with those from multiple linear regression.

I've chosen the ==number of hidden neurons==(n_hidden) as the hyperparameter of my ELM. Also, I did the 5-fold cross validation to select the best n_hidden as 5. Here's my output result of cross validation(each item in CV rmse results means n_hidden vs. corresponding rmse):

```
CV rmse results: {5: 490.816, 10: 526.899, 20: 1781.641, 50: 34675.982, 100:
15026.577}
Best n_hidden: 5
```

Finally, let's compare the rmse and the correlation coefficient between the ELM and MLR:

```
rmse_ELM_ensembles=492.876, rmse_MLR=519.302
corr_ELM_ensembles=0.493, corr_MLR=0.37
```

It's easy to observe that rmse is lower and corr is higher for ELM, indicating it's a better model than the LMR.

Problem1 Code

2. Complete Exercise 8.1 in Hsieh's book. Please tune the learning rate for the MLP NN model.

8.1

Let $y_{\text{signal}} = \sin(2\pi x)$ be the signal, with the random variable x uniformly distributed in the interval $[-1,\ 1]$. The y data are generated by adding Gaussian noise to y_{signal}, where the Gaussian noise has the same standard deviation as y_{signal}. Generate 200 training data points and 1,000 test data points. Fit an ensemble of MLP NN models with one hidden layer to the (x, y) training data, using some form of regularization (e.g. early stopping by reserving 15% of the training data for validation and training with 85% of the training data). Vary the number of hidden neurons (at 2, 3, 4, 5, 6, 7 and 8) and the ensemble size (at 25, 50 and 100). From the test data, determine (i) the average RMSE of an individual model's prediction of y and (ii) the RMSE of the ensemble average prediction of y.

(a) Are the values in (ii) smaller than the corresponding values in (i)?

(b) Do the ensemble average using 100 ensemble members outperform that using 25 ensemble members?

(c) Repeat with the Gaussian noise at 0.5 and 2 times the standard deviation of y_{signal}. Do your answers in (a) and (b) still hold at the lower and higher noise levels?

I used `Keras` to build and train NN models, in particular, using the `callbacks.EarlyStopping` utility helped the tuning of the learning rate. Also, the training of NN models take time, so I split the workflow into 2 steps, namely train/pred and evaluation by saving the predicting values as `npy` file. Finally here is my full resulting table:

note:

- avg_rmse: in an ensemble, for each y_pred, calculate rmse, and then calculate avg(rmse)
- ensemble_emse: in an ensemble, calculate avg(y_pred), and then calculate its rmse

	sigma	n_hidden	n_ensemble	avg_rmse	ensemble_rmse	best_lr
0	noise-0.5sigma	2	25	23.3782	23.2505	0.000259294
1	noise-0.5sigma	2	50	21.8022	20.8957	0.000259294
2	noise-0.5sigma	2	100	26.4893	24.8314	1.61026e-05
3	noise-0.5sigma	3	25	24.4425	24.1905	3.56225e-05
4	noise-0.5sigma	3	50	24.4663	24	5.29832e-05
5	noise-0.5sigma	3	100	17.8284	14.9309	0.1
6	noise-0.5sigma	4	25	23.8701	23.8362	2.39503e-05
7	noise-0.5sigma	4	50	16.9993	15.2766	0.00923671
8	noise-0.5sigma	4	100	26.538	25.0681	0.00621017

	sigma	n_hidden	n_ensemble	avg_rmse	ensemble_rmse	best_lr
9	noise-0.5sigma	5	25	25.3555	24.6693	7.27895e-06
10	noise-0.5sigma	5	50	16.5668	16.4842	0.0452035
11	noise-0.5sigma	5	100	24.4073	24.1205	0.00011721
12	noise-0.5sigma	6	25	24.0951	23.9904	2.39503e-05
13	noise-0.5sigma	6	50	12.7082	12.6402	0.0672336
14	noise-0.5sigma	6	100	24.1184	23.9754	1e-06
15	noise-0.5sigma	7	25	23.8388	23.8353	3.56225e-05
16	noise-0.5sigma	7	50	16.9012	16.1963	0.000259294
17	noise-0.5sigma	7	100	25.7091	24.3466	0.00280722
18	noise-0.5sigma	8	25	25.4185	24.2305	1.48735e-06
19	noise-0.5sigma	8	50	17.1544	17.1466	0.0137382
20	noise-0.5sigma	8	100	13.0915	12.9384	0.1
21	noise-1sigma	2	25	33.3813	31.3964	0.000174333
22	noise-1sigma	2	50	27.9583	25.6886	0.00280722
23	noise-1sigma	2	100	26.031	25.9017	0.0137382
24	noise-1sigma	3	25	29.7305	29.5679	0.000385662
25	noise-1sigma	3	50	30.8486	30.8301	2.39503e-05
26	noise-1sigma	3	100	30.81	30.7858	1.48735e-06
27	noise-1sigma	4	25	31.0964	31.094	3.29034e-06
28	noise-1sigma	4	50	23.371	23.3494	0.1
29	noise-1sigma	4	100	26.5657	25.5499	0.00188739
30	noise-1sigma	5	25	32.2665	31.7617	0.00280722
31	noise-1sigma	5	50	31.8778	31.3293	1.08264e-05
32	noise-1sigma	5	100	31.1727	31.153	3.56225e-05
33	noise-1sigma	6	25	25.9683	25.5905	0.00126896
34	noise-1sigma	6	50	26.9496	25.8542	0.00188739
35	noise-1sigma	6	100	26.6801	26.4703	0.1
36	noise-1sigma	7	25	23.5906	23.4312	0.1
37	noise-1sigma	7	50	30.9757	30.973	0.030392
38	noise-1sigma	7	100	23.4616	23.4177	1.08264e-05

	sigma	n_hidden	n_ensemble	avg_rmse	ensemble_rmse	best_lr
39	noise-1sigma	8	25	26.0009	25.9177	0.00126896
40	noise-1sigma	8	50	30.262	30.2165	0.0672336
41	noise-1sigma	8	100	31.0681	31.0671	7.88046e-05
42	noise-2sigma	2	25	48.3115	47.804	0.00126896
43	noise-2sigma	2	50	48.0483	46.7206	0.0452035
44	noise-2sigma	2	100	51.2101	49.8417	2.21222e-06
45	noise-2sigma	3	25	50.4188	50.0577	1.61026e-05
46	noise-2sigma	3	50	47.0717	45.7904	0.00417532
47	noise-2sigma	3	100	49.8791	49.7953	4.8939e-06
48	noise-2sigma	4	25	50.3468	50.0683	0.00188739
49	noise-2sigma	4	50	49.7143	49.7104	0.0204336
50	noise-2sigma	4	100	51.955	50.3372	0.1
51	noise-2sigma	5	25	49.8459	49.7974	0.000259294
52	noise-2sigma	5	50	49.8696	49.7575	0.1
53	noise-2sigma	5	100	51.0832	49.8741	0.0672336
54	noise-2sigma	6	25	49.5725	49.5682	0.1
55	noise-2sigma	6	50	47.1561	47.0253	0.030392
56	noise-2sigma	6	100	49.7303	49.7278	0.1
57	noise-2sigma	7	25	47.0183	46.9967	0.030392
58	noise-2sigma	7	50	47.179	47.0493	0.0452035
59	noise-2sigma	7	100	51.4086	50.1199	0.0204336
60	noise-2sigma	8	25	47.2009	46.9837	0.1
61	noise-2sigma	8	50	46.8958	46.8475	0.1
62	noise-2sigma	8	100	47.1671	46.9541	7.88046e-05

Some discussions

1. Answering (a): Yes, `ensemble_rmse` are almost always less than `avg_rmse`, indicating that ensemble method does reduce prediction error. Hence it's more meaningful to talk about the `ensemble_rmse`.
2. Answering (b): For `0.5sigma`, `n_ensemble`=100 is not better than `n_ensemble`=25, as indicating by the row index=0(25 ensemble_rmse=23.2505) versus row index=2(100 ensemble_rmse=24.8314)
3. Answering (c): (a), (b) still hold within different noise levels.
4. There seems no clear pattern in the `best_lr` with respect to the number of hidden neurons.

5. `ensemble_rmse` increases as noise level increases, which is fairly reasonable.

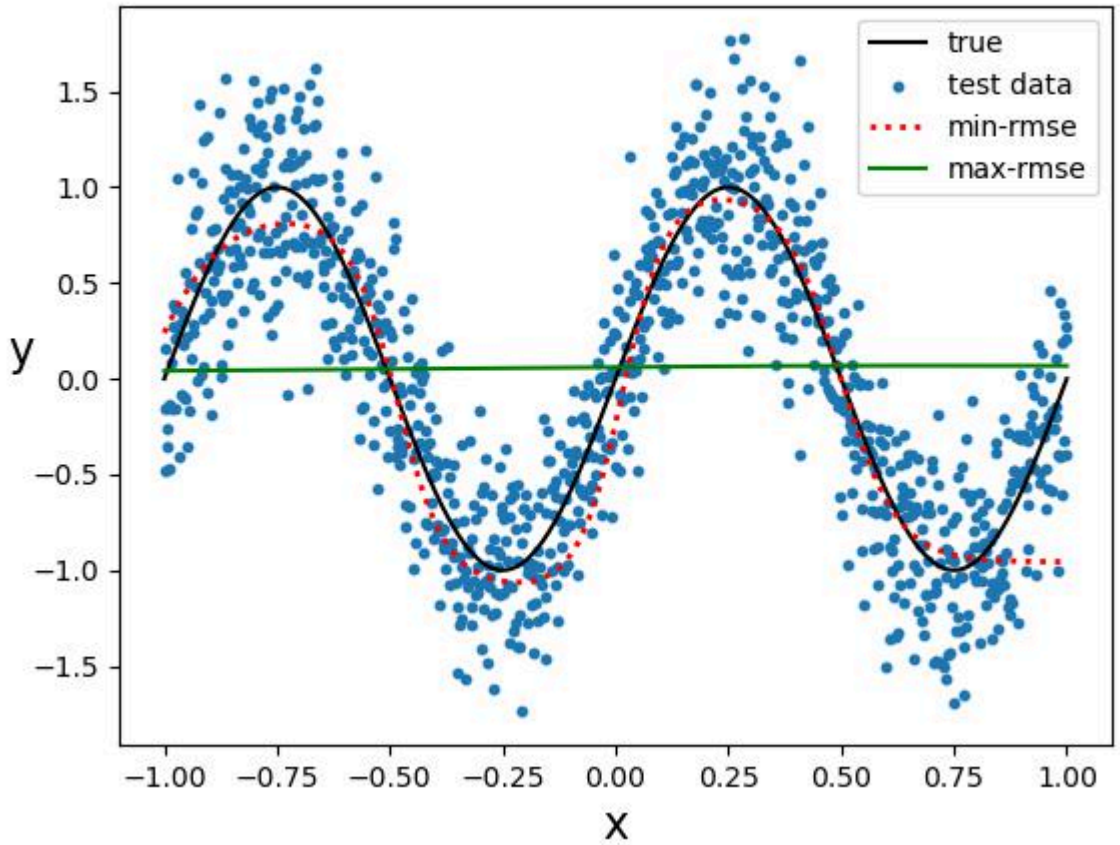
Problem 2 train/pred code

Problem 2 evaluation code

3. Visualize the regression results of Exercise 8.1 at least for the case with the Gaussian noise at 0.5 times the standard deviation of y_{signal}

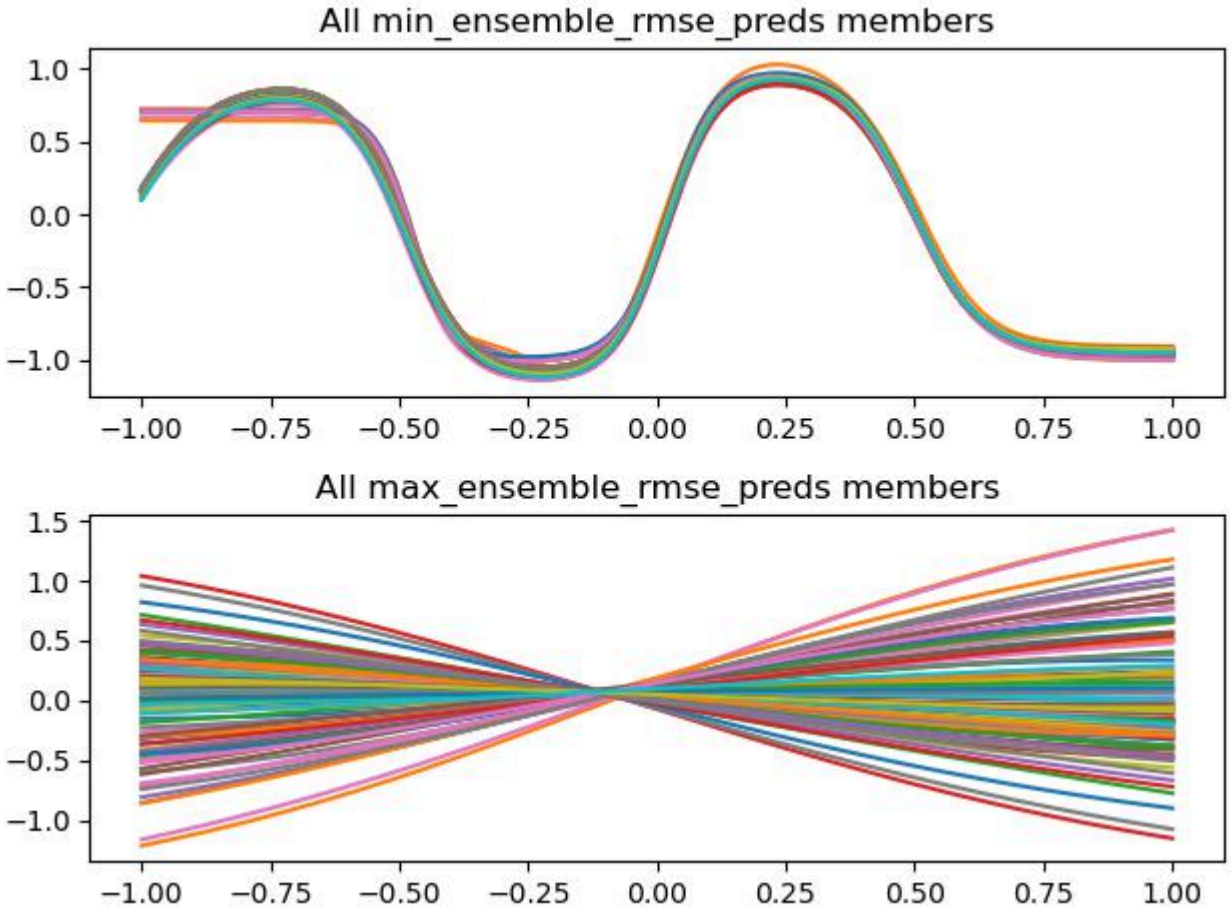
Since there are many combinations with respect to `n_hiddens` and `n_ensembles`, I've chosen the ones with highest/lowest `ensemble_rmse`, and plotted its ensemble mean results versus the test data and the true relation as below:

min `ensemble_rmse`: `n_hiddens`=6, `n_ensembles`=50 max `ensemble_rmse`: `n_hiddens`=4, `n_ensembles`=100



It's easy to see for the case of max `ensemble_rmse`, the result looks like a straight line, indicating this model did not learn any non linear feature of the data, maybe it needed more training epochs!

To investigate further, I've plotted all members for those two cases as below:



None of the member in the case of max `ensemble_rmse` capture non linearity!

Problem 3 code